# Angles & Parallel Lines Maze ANSWER KEY

Start in the top-left room. Solve the problem, find the door with your answer, and go through it. Show your work on your answer grid.

**START**

**1** Vertical angles. Find x.

$(3x + 10)°$ and $70°$

$x = 20$

**2** Find x.

Linear pair

$2x°$ and $(x + 30)°$

$x = 50$

**3** Lines ∥. Find x.

Corresponding ∠s

$(5x - 15)°$, $(3x + 21)°$

$x = 3$

Lines ∥. Find the angle measure.

Alternate interior ∠s

$(9x - 20)°$, $(5x + 24)°$

$x = \frac{80}{3}$

$x = 60$

$x = 30$

$x = 70$

$x = 18$

$x = -18$

11°

Lines ∥. Find the angle measure.

Corresponding ∠s

$(8x - 2)°$, $(6x + 14)°$

$x = 41$

**5** Lines ∥. Find x.

Same-side interior ∠s

$(2x + 20)°$, $(3x + 5)°$

**66°**

**4** Lines ∥. Find the angle measure.

Alternate interior ∠s

$(4x + 6)°$, $(6x - 24)°$

96°

Find x.

Vertical angles

$(2x + 41)°$, $(4x - 5)°$

113°

$x = 31$

$x = 15$

114°

15°

68°

$x = -23$

**6** Lines ∥. Find the angle measure.

Alternate exterior ∠s

$(7x - 8)°$, $(5x + 22)°$

**8** Find the angle measure.

Vertical angles

$(5x - 12)°$, $(3x + 16)°$

**9** Lines ∥. Find the larger ∠.

Same-side interior ∠s

$(6x + 4)°$, $(4x - 4)°$

Lines ∥. Find x.

Same-side interior ∠s

$(x + 15)°$, $(2x + 21)°$

83°

**58°**

90°

$x = 24$

**97°**

**120°**

14°

**112°**

18°

$x = -6$

Find x.

Linear pair

$(5x + 3)°$, $(3x + 9)°$

**7** Linear pair. Find m∠1.

$m\angle 1 = (4x + 12)°$

$m\angle 2 = (2x + 6)°$

**10** Lines ∥. Find x.

Same-side interior ∠s

$3(x + 10)°$, $2(x - 5)°$

27°

60°

$x = 32$

**FINISH**

You made it!

# Angles & Parallel Lines Maze: Answer Grid **ANSWER KEY**

Write each answer in the order you found it, and show your work for that room.

| | | |
|---|---|---|
| **1** | Answer: **$x = 20$** | $3x + 10 = 70 \rightarrow 3x = 60 \rightarrow x = 20$ |
| **2** | Answer: **$x = 50$** | $2x + x + 30 = 180 \rightarrow 3x = 150 \rightarrow x = 50$ |
| **3** | Answer: **$x = 18$** | $5x - 15 = 3x + 21 \rightarrow 2x = 36 \rightarrow x = 18$ |
| **4** | Answer: **66°** | $4x + 6 = 6x - 24 \rightarrow x = 15 \rightarrow 4(15) + 6 = 66$ |
| **5** | Answer: **$x = 31$** | $5x + 25 = 180 \rightarrow 5x = 155 \rightarrow x = 31$ |
| **6** | Answer: **97°** | $7x - 8 = 5x + 22 \rightarrow x = 15 \rightarrow 7(15) - 8 = 97$ |
| **7** | Answer: **120°** | $6x + 18 = 180 \rightarrow x = 27 \rightarrow m\angle 1 = 4(27) + 12 = 120$ |
| **8** | Answer: **58°** | $5x - 12 = 3x + 16 \rightarrow x = 14 \rightarrow 5(14) - 12 = 58$ |
| **9** | Answer: **112°** | $10x = 180 \rightarrow x = 18 \rightarrow 6(18) + 4 = 112$ |
| **10** | Answer: **$x = 32$** | $3x + 30 + 2x - 10 = 180 \rightarrow 5x + 20 = 180 \rightarrow x = 32$ |